Idon-Lafia: PHC Triage

YoungDataPro Datathon 2026, Health Track · All outcome data in this project is synthetic and labelled "Sample data".

1. The problem we chose

Nigeria has about 31,000 primary health centres (PHCs), most people's first contact with the health system. Facilities are meant to report medicine, vaccine, staffing and service data upward (facility → LGA → state → DHIS2), but reports arrive late, incomplete or not at all. So officers and planners are guessing, and a medicine can run out with nobody above the facility knowing until a patient arrives and there is none.

Evidence of the gap: - The NBS **2023 National Health Facility Survey** sampled 3,330 facilities. It found that on average only **34.3% of essential drugs were in stock and in date in primary health facilities**. This is the share of essential drugs present, not the share of facilities. - State figures ranged from about **54% in Delta to about 14% in Kebbi**.

The capacity gap: an LGA officer looks after roughly **40 facilities**, our estimate from about 31,800 PHCs across 774 LGAs, but can only visit a few each week.

From the brief we took the **risk-scoring** and **anomaly-detection** directions: *flag which facilities are at the highest risk of running short of medicines, vaccines, staff or critical supplies before it becomes a crisis*, and treat **reporting gaps themselves as a warning sign**. The question our tool answers is:

> **"Which facilities should I check first this week, and why?"**

We sit on top of DHIS2 and only read from it. We never replace it or write back to it.

2. Our approach

1. **Data.** Real facility-level DHIS2 data is not public, so we built a simulator that places synthetic monthly reports on **real GRID3 facility locations** in Oyo, Kano and Gombe (1,537 facilities, 37 LGAs, 48 months), calibrated to NMIS averages. Each facility has a hidden "true" state (stock, fridge, staff, closure) that the model never sees. The model only sees what a DHIS2-style system would: service counts, and whether each report was on time, late or missing. Eight stress-test worlds (weak link, high noise, insecure, many chronic poor reporters, and others) check that results are not tuned to one set of assumptions.
2. **Three models** (LightGBM):
3. **Expected report:** how much activity this facility *should* have reported this month. A large gap is a warning sign.
4. **Silence type:** when a facility goes quiet, why? The options are a data-entry delay, a chronic poor reporter, a real problem, an LGA-wide backlog, or closure.
5. **Trouble forecast:** a calibrated chance of serious trouble in the next 1–2 months, overall and per failure type (medicine, vaccine, staff, closed), with an uncertainty band from a 5-model ensemble.
6. **Priority** = trouble chance × **impact** (population served, and the distance to the next facility). A remote clinic serving many people outranks a small one in town with similar risk. Facilities are tagged HIGH, MEDIUM or LOW.
7. **Explanations.** SHAP values are grouped into plain reasons (reporting/silence, medicine, vaccine/cold chain, staff, distance, security, season). Each flag comes with a sentence and a suggested action.
8. **Honest testing.** We trained on months 1–30, tuned on 31–39 and tested on 40–48, with a 2-month gap so labels never leak. A further 20% of LGAs were held out of training entirely. We compared against simple rules an officer could use without us.

3. Results (synthetic data)

Measure	Model	"Silent 2+ months" rule	Random
Right in the top 5 per LGA	**82%**	49%	31%
LGAs never seen in training	**86%**	52%	–
Rural vs urban precision	83% vs 82%	–	–
All 8 stress-test worlds	about 80%	about 48%	about 31%

Calibration: Brier 0.157, AUC 0.786. An officer who visits our top 5 finds real trouble about 4 times in 5, against about 1 in 2 with the simple rule.

Known weakness: in the very first month of new trouble, the simple rule catches more (15% vs 5%). Our model is better at ranking facilities that are already sliding than at spotting month one.

4. What we built

Dashboard (Firebase Hosting). The pages work on a phone and can be printed. - **Officer list:** - Pick your LGA and how many visits you can make this week, then get a ranked list. Each card shows the tag, trouble chance, reason and suggested action. - The list is split into *To check*, *Waiting for an answer* and *Follow-up*. - LGA-wide alerts flag when many facilities go quiet together, which usually means a data-entry backlog rather than many failures. - Export the list as CSV or open a printable version. - **Facility page:** - A reporting-history chart (expected vs reported). - "Why it is flagged" bars, the silence type, and tiles for days since last report, staff, cold chain, connectivity and distance. - The suggested action. - **State overview:** totals, LGAs ranked by flagged facilities, cluster alerts, a breakdown of why facilities are quiet, and the full validation panel.

Feedback loop. - The officer records the result of each check: *Checking*, *Confirmed*, *Cleared*, *Just late* or *Could not verify*. They also record how it was checked (phone, visit or committee SMS) and the problem type. - Cleared and just-late facilities are snoozed for 45 days. A facility comes back early if its risk jumps by 0.2 or more. - Three or more "just late" answers in one LGA alert the data office about a backlog. - An officer can override a flag with a reason, for 30 days. - Every answer is saved with the score and model version at the time, so it can become a training label. A simulated feedback loop shows retraining on these answers keeps precision high.

WhatsApp (Meta WhatsApp Business Cloud API). - **Automatic urgent alerts:** - An officer saves their number once. They immediately get a WhatsApp message listing every HIGH-priority facility in their LGA, with links to each facility page. - The backend re-checks every hour and after a retrain. It only messages when a facility *newly* becomes HIGH, so there are no repeats. - Every send is logged on the panel, and failed sends are shown in red. - **Two-way:** the officer can text the business number. "START alerts " subscribes and "STOP" unsubscribes. - **Manual messages:** a weekly brief to the officer, and a report reminder or YES/NO fridge check to a facility in-charge. Each is previewed first and sent only after a human confirms.

Phone calls (BimpeAI voice agent). - WhatsApp needs mobile data, so officers can also tick **"Also phone me"** to be reached on any phone. - A BimpeAI voice agent phones the officer on the number they saved whenever an issue becomes urgent, telling them which facilities need checking and why. It can repeat the message and answer simple questions about the alert. - Calls follow the same rules as WhatsApp: immediately on subscribe, then only for newly urgent facilities. There is also a **Call me now** button. - On the Facility page, an officer can phone the in-charge with a spoken report reminder or fridge check.

Safety choices. - The phone numbers in the dataset are fake samples. They are shown only as hints and never dialled or messaged. - Only numbers a person types in are contacted. - The reminder button is disabled for a facility that reported on time. - Secrets live in `.env` and are never committed. - All data is labelled synthetic.

5. How it runs

- **Backend:** FastAPI (Python) on port 5005. It serves the API and also the dashboard.
- **Public access:** `start_tunnel.ps1` opens a Cloudflare quick tunnel, points the dashboard at it and redeploys to Firebase.
- **WhatsApp:** Meta's webhook calls `…/whatsapp` on the backend.

Code layout: - `phc_triage/simulator/` : builds the synthetic world. - `phc_triage/ml/` : models, training and evaluation. - `phc_triage/backend/` : the API. - `alerts.py` : urgent alerts. - `whatsapp.py` : WhatsApp sending. - `calls.py` : BimpeAI phone calls. - `phc_triage/dashboard/` : the web app.

6. Assumptions and next steps

Assumptions. - Late and missing reports carry signal about real trouble. The c-sweep shows the model still beats the simple rules even when that link is weak. - The population and distance to the next facility are reasonable proxies for the impact of a failure. - Officers will record check results honestly. We simulated officers with 70–100% accuracy.

Next steps. - Pilot on real DHIS2 exports from one LGA. - Officer login. - An in-charge view. - Offline use.